

Dachuan Song

Fairfax, Virginia, 22030 | dsong25@gmu.edu | (571) 530-8724

Google Scholar | GitHub | LinkedIn | Personal Website

Education

George Mason University, Fairfax, Virginia, USA Jan 2024 – Present

College of Engineering and Computing

Ph.D. in Electrical and Computer Engineering, Current GPA: 3.91

Advisor: Prof. Xuan Wang

- **Major Courses:** *Linear Systems, Random Processes, Kalman Filtering, Distributed Control and Optimization, System Identification, Advanced learning from data*

University of Southampton, Southampton, United Kingdom

Sept 2022 – Sept 2023

School of Electronics and Computer Science

M.Sc. in Computer Science, GPA: 3.8 (Top 3%)

- **Major Courses:** *Advanced Machine Learning, Advanced Deep Learning, Computer Vision, Bayesian, Active & Reinforcement Learning*

Xinjiang University, Xinjiang, China

Sept 2017 – Jun 2021

School of Software

B.Eng. in Software Engineering, GPA: 3.4 (Top 5%)

Publications

CONFERENCE PAPERS AND PREPRINTS:

Elastic Spectral State Space Models for Train-Once Budgeted Inference 2026

Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang

<https://arxiv.org/abs/2601.22488>

Accepted to *NeurIPS 2026 Main Track (Poster)*; proceedings forthcoming. *arXiv:2601.22488, 2026.*

Elastic Selective Spectral Hybrids for Train-Once, Export-Many Budgeted Inference Sep 2026

Dachuan Song, Chuchu Chen, Xuan Wang

<https://arxiv.org/abs/2609.32486>

Mergeable Model-Side Aggregation States for Long-Context Language Models Jul 2026

Dachuan Song, Junyu Yin, Zechen Hu, Xuan Wang

<https://arxiv.org/abs/2607.26448>

Causality-based Subject and Task Fingerprints using fMRI Time-series Data Nov 2024

Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang

<https://doi.org/10.1145/3698587.3701342>

ACM BCB '24, Article 13, pp. 1–10 (Oral Presentation).

JOURNAL ARTICLES:

Reconstructing brain causal dynamics for subject and task fingerprints using fMRI time-series data Oct 2025

Dachuan Song, Li Shen, Duy Duong-Tran, Xuan Wang

<https://doi.org/10.1007/s13755-025-00388-w>

Health Information Science and Systems 13, Article 70 (2025).

Research Projects

Addressable Memory for Recurrent Language Models

Sep 2026 – Present

- Developed delta-rule language models with exact-address memory, combining n-gram indexing with bounded candidate retrieval for long-context factual recall.
- At 135M parameters and 1B training tokens, achieved 25.34 validation perplexity versus 25.82 for a parameter-matched KDA baseline, with 99.0% token-level needle-recall accuracy at a 14K-token retrieval distance in 16K-token contexts.
- Designed learned address-generation and source-supervision mechanisms, with controlled ablations and causality tests to identify failures in addressing, selection, and readout.

Elastic Selective Spectral Hybrids for Budgeted Inference

Jul 2026 – Sep 2026

- Gave each spectral channel a recurrent state with input-dependent decay and write/read gates, so it can retain relevant context and forget distractions. Sliding-window attention handles recent tokens.
- Trained channel count and feed-forward width as capacity tiers, producing compact exports from one checkpoint. At 1.53B parameters, fused batch-one decoding took 1.37 ms/token on a B300, 2.14–2.80 $\times$ faster than the tested Mamba-2 and Mamba-3 implementations at matched parameter counts.

Elastic Spectral State Space Models for Train-Once Budgeted Inference

Jan 2026 – May 2026

- One training run yields independently deployable models for different budgets. Direct truncation removes ordered Hankel channels and their parameters from each smaller export.
- Input-adaptive gates tune channel contributions; budget dropout trains every retained prefix. Across language, audio, long-range, and offline reinforcement-learning tasks, quality changes smoothly as the budget shrinks.

Mergeable Model-Side Aggregation States for Long-Context Language Models

Jul 2026 – Aug 2026

- Added a 2 KiB mergeable HyperLogLog state beside a frozen language model to track distinct counts and set evidence across context chunks.
- Reached 99.2% accuracy on 3,969 aggregate-then-reason tasks with Gemma 4 (31B), without rereading the full history.

Machine Learning for Brain Activity Patterns using EEG (Master's Project)

Apr 2023 – Oct 2023

- Combined CNN spatial features with LSTM temporal modeling for EEG brain-pattern recognition.
- Used autoencoder pretraining, attention, and transfer learning to extract features from limited EEG data.

Internship Experience

Machine Learning Engineer Intern, Optosurgical

May 2026 – Aug 2026

- Recognized surgical phases and instrument actions in video, providing procedure-state signals to downstream systems.
- Compressed and fine-tuned Depth Anything 3 to 29% of its original size while retaining near-full depth-estimation performance.

Technical Skills

- **Architectures & Inference:** spectral and selective state space models, recurrent memory, sliding-window attention, long-context aggregation, multi-budget model export.
- **ML Engineering:** PyTorch, CUDA, FFT convolution, multi-GPU training, model compression, fused recurrent decoding, reproducible evaluation.
- **Modeling & Control:** linear systems, Kalman filtering, system identification, LQR/MPC, autoregressive models.